

PHISHING EMAIL DETECTION & AWARENESS REPORT

Task 2 – Phishing Email Detection & Awareness System

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Task Description: Analysis of phishing email samples to identify phishing indicators, classify email risks, assess potential business impact, and provide security awareness recommendations and preventive measures.

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Executive Summary

Phishing is an ever-growing threat to individuals, businesses, and government agencies worldwide. Cybercriminals use phishing emails to trick target users into disclosing personal information, clicking on malicious links, downloading malicious attachments, and providing login credentials to access their accounts. Mostly, phishing attackers use social engineering techniques to gain unauthorized access to their targets' systems and devices instead of traditional hacking methods. Therefore, it is essential for members of an organization to be aware of the possible phishing attack signs and respond appropriately to protect their personal and organizational data. This paper will discuss a report on phishing email detection and awareness assessment conducted via testing on public phishing emails for research and educational purposes.

The phishing email detection and awareness report aims to detect and assess the level of danger of phishing emails using public phishing emails as a sample. The main purpose of the phishing awareness assessment is to consider various aspects of emails, detect phishing indicators, determine the level of danger, and provide warnings on how to avoid similar phishing emails. The assessment was performed in a responsible and ethical manner, ensuring that there was no interaction with phishing links or attachments.

The report contains the analysis of 5 email samples, including phishing, suspicious, and safe email samples. Each sample was considered and evaluated to identify suspicious patterns, including sender address, subject line, phishing indicators, presence of links and attachments, and suspicious language. Moreover, based on the detected phishing indicators, each email was categorized as Phishing, Suspicious, or Safe to determine the level of possible danger. During the assessment, the suspicious patterns that can indicate phishing were detected, describing why the email was classified as phishing, suspicious, or safe. Furthermore, the report highlighted the possible impact of phishing emails on the target users and organizations, outlining the potential damage, including the loss of personal and organizational data, disruption of normal operations, and economic loss. In this report, apart from the analysis of phishing emails, the phishing prevention and awareness recommendations are also provided to increase the security awareness of users and ensure that they know how to act if they suspect phishing attempts.

In conclusion, this paper provided the phishing email detection and awareness report based on the assessment of public phishing emails. The assessment showed that phishing attackers used social engineering techniques to manipulate victims and trick them into providing personal information and sensitive data by impersonating well-known and reliable senders. The report demonstrated that phishing emails could have a significant impact on the target users, ranging from stealing their personal information to disrupting the operation of a business. This report also showed that it is essential for organizations to provide cybersecurity awareness training to their employees to ensure that they know how to detect phishing emails and respond to them appropriately.

Introduction

Phishing is one of the most common and fast-evolving security threats that modern individuals and businesses face. This threat is classified as a social engineering technique that consists of criminals sending out emails designed to look like they are coming from legitimate organizations like banks, tech companies, shippers, or other corporate entities. The goal of phishing emails is usually to manipulate individuals into disclosing personal information or interacting with malicious content, such as attachments or links.

Since phishing is a social engineering threat, it primarily targets the psychological aspects of individuals, making it one of the hardest cyber security issues to address. In many ways, phishing emails are difficult to differentiate from legitimate ones since they often appear as if they have been sent by reliable organizations. However, in most cases, phishing emails ask the recipient to do something they would not normally do, such as share passwords or update account details or provide the location of a courier that the company would typically provide. With the evolution of phishing emails, businesses are tasked with ensuring that their employees are aware of how to deal with them.

I offer a phishing email detection & awareness assessment service based on public information about phishing emails for educational and other informational purposes. As part of the service, several samples of phishing emails had been reviewed to help identify phishing attempts and raise awareness against them. During the assessment, I analyzed phishing emails to determine the indicators of phishing such as suspicious email addresses, generic greetings, urgency, malicious links, and attachments, and requests for confidential information. During the assessment, emails were reviewed and categorized as Phishing, Suspicious, or Safe.

In most cases, the main goal of phishing emails is to trick individuals into disclosing personal information or interacting with dangerous links or attachments. Therefore, this report also aims to educate the reader about phishing emails by discussing the unique ways that phishing campaigns operate in simple words, while also providing prevention measures and awareness recommendations. This assessment service helps raise awareness about phishing emails, which ultimately benefits businesses by helping them enhance their cyber security awareness and reduce phishing-induced threats.

Additionally, the assessment results help understand the significance of taking a more active role in ensuring email security. It is essential to remain cautious about the source of incoming emails, check their content, address any suspicious activity, or report it to the relevant authorities. The combination of ongoing security awareness training, email filtering solutions, and multifactor authentication can remarkably minimize the success rate of phishing attempts. It is critical to conduct regular cybersecurity training for all members of an organization to prevent sensitive data breaches, retain customer trust, and avoid adverse publicity. In general, the overall assessment results demonstrate that cybersecurity is not just about having the proper technological defenses but rather about ensuring everyone is educated about the possible threats and can adequately respond to them, such as recognizing phishing emails.

Objectives

The main purpose of the Phishing Email Detection & Awareness Assessment is to analyze phishing email samples and highlight the differences that can help distinguish phishing emails from legitimate ones. This assessment aims to raise awareness about phishing, explain what phishing attempts are, and how to protect oneself from them. Since phishing relies mostly on human error, the awareness level has a significant influence on the possibility of a successful attack. Another objective of the assessment is to go through the elements of each email and highlight the similarities and differences.

Among other goals, the assessment also aims to explain the significance of classification and risk levels for emails. For each email, an in-depth analysis is made to spot the features that can help distinguish phishing from legitimate emails. Moreover, each email is classified into Phishing, Suspicious, and Safe categories. Classification allows an adequate perception of the risk level of each email. In addition, the impact of the phishing emails on the business and the possible consequences of an attack are discussed. Furthermore, the analysis provides the recommendations for protection from phishing.

Another goal of the assessment is to raise awareness of the essential aspects of emails that should be examined closely when attempting to identify phishing. The assessment goes through all the elements of the email, such as the sender's address, subject, email body, hyperlinks, attached files, and overall design. The main features of phishing emails, which might differ from legitimate ones, include sender's address, subject, and body, red flags, hyperlinks, attachments, and other details. By detecting and analyzing these details, it is possible to determine if an email is phishing, suspicious, or safe.

The aim of the assessment is also to explain the significance of analyzing email traffic and understanding the risks in business. This objective is achieved by observing the potential risks and threats posed by phishing emails. These risks include confidential information exposure, identity theft, financial theft, hijacking, malware introduction, disruption of business operations, and damage to the company's reputation. All these risks are presented in simple and clear language so that all employees can easily understand them. Furthermore, the objective of the analysis is to provide recommendations that may offer protection from phishing. This goal is reached by providing the list of suggested measures for protection from phishing, such as checking the sender's address, examining links and attachments, and raising awareness. The recommendations should help the employees improve the security posture of the company's email system. Ultimately, the assessment should demonstrate proper and ethical use of examples of phishing emails for educational purposes. This objective is achieved by highlighting the importance of phishing assessments for businesses. The assessment is an example of how phishing emails can be analyzed in order to improve the safety of the email system. During the assessment, actual links were not opened, files were not executed, or any other actions that potentially could compromise the system security were taken.

Scope

The scope of this Phishing Email Detection & Awareness Assessment is to analyze phishing email samples using standard and non-malicious ways for awareness and educational purposes. The analysis focuses on the identification of phishing email characteristics, the impact assessment of phishing attempts on the organization security, and the establishment of email security best practices that can help users avoid phishing scams and respond appropriately to such incidents.

The phishing email analysis covered five different email samples (with multiple emails included per each category), including Phishing, Suspicious, and Safe categories. For each email, the sender's email address, subject, body, links, attachments, formatting, and language were examined and reviewed. Specific characteristics of phishing emails, including suspicious sender, suspicious subject, phishing language, phishing attachments, phishing links, fake sender's address, and misspelled words were used for risk assessment and categorization.

In terms of methodology, the email headers were analyzed using public email header analysis tools for additional inspection for each email's attributes as needed for a comprehensive inspection. Browser inspection techniques were used to analyze the phishing emails for malicious links, scripts, attachments, and other suspicious content. No malicious activity was conducted by the team during the scope of this task, as all analysis was done in a safe and segregated environment for cybersecurity assessment and research purposes only.

Moreover, the scope of this phishing assessment covers the analysis of business impact that phishing attempts can have on the company. This report reviews and expands on different business impacts that phishing emails can cause, from stealing user credentials on company systems and exfiltrating data to disrupting normal business operations and damaging the company's reputation. In addition, recommended user actions are discussed, including steps users can and cannot take when they suspect that they received a phishing email and general user awareness considerations. The scope of this assessment does not cover malware analysis, penetration testing, or exploitation attempts of any vulnerabilities, as no malicious links, attachments, or exploit payloads were interacted with during this assessment. The report's findings and recommendations can provide the company with significant insight and awareness into the threat of phishing email security incidents that the organization faces, as well as build the necessary awareness to mitigate and reduce the risks associated with such incidents. This assessment covers a limited scope of analyses that can be conducted on phishing email samples, primarily centered around inspection and risk assessment for educational and awareness-building purposes.

Tools Used

The phishing email examples were gathered from public educational resources to ensure safety. Gmail Message Header Analyzer and MXToolbox were utilized to scrutinize the message headers to determine the sender’s data and email traffic route, if necessary. Mozilla Firefox and Browser Developer Tools were deployed to manually check the message’s links and contents for malicious scripts, attachments, or threats. The report was designed using Canva, and its contents were composed using Microsoft Word/Gmail Docs. GitHub was used to store the project’s data and reports.

Tool	Purpose
Public Phishing Email Samples	Collected phishing, suspicious, and legitimate email samples for analysis.
Google Message Header Analyzer	Analyzed email headers to examine sender information and email routing details.
MXToolbox Email Header Analyzer	Verified email header information and assisted in identifying suspicious email characteristics.
Mozilla Firefox	Safely viewed email samples and inspected links without interacting with malicious content.
Browser Developer Tools	Examined URLs, webpage elements, and network requests for analysis purposes.
Canva	Designed and formatted the professional phishing detection and awareness report.
Microsoft Word / Google Docs	Prepared and documented the assessment findings before converting the report to PDF.
GitHub	Maintained a public repository containing the report, evidence, and project documentation.

Table 1- Tools and Purpose Summary

Assessment Methodology

The phishing email detection and awareness assessment was carried out by following a particular approach to analyze the email samples and detect phishing attempts. The methodology was aimed at increasing cybersecurity awareness by reviewing publicly available phishing email samples. The methodology was based on ethical principles and allowed detecting phishing attempts by analyzing the phishing email samples without clicking on the links and opening attachments. Each step of the methodology was developed to ensure a safe analysis of the phishing email samples and prepare a phishing email detection report according to the industry standard.

The first step in the methodology was to collect publicly available email samples that could represent the different categories of emails, such as Phishing, Suspicious, and Safe. In order to do that, the emails that could be used as examples of the Phishing category included Fake Fax Message, Fake Voice Message, Microsoft Password Expiration, Suspicious Account Activity, and Safe Account Notification. Each category had several samples that could be used to compare the differences between the phishing emails and safe emails.

The second step was to analyze each email sample by reviewing the email header, subject, body, formatting, hyperlinks, attachments, and other elements. Each email was analyzed in terms of the presence of phishing indicators, such as sender address, subject line, message content, suspicious formatting, links, attachments, and other elements. Particular attention was paid to the sender address, subject line, and message content, as they normally indicate the presence of phishing attempts.

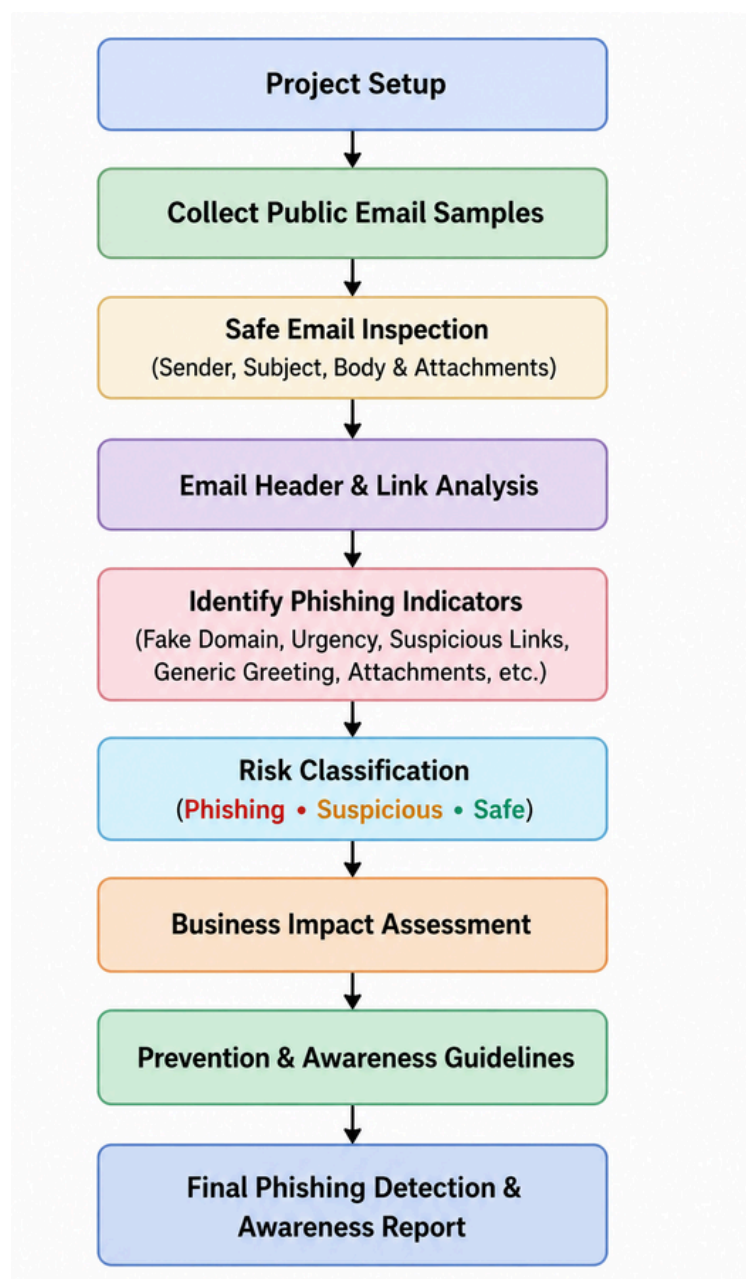
When analyzing email headers, the publicly available email header analysis tools were used to check the sender information and email route. Besides, the browser's inspection tool was used to review the email content and check the links for malicious scripts. Particular attention was paid to the links and attachments in the email, as they normally contain the main indicators of phishing attempts. At the same time, no links were opened, and no attachments were opened for analysis in order to ensure the safety of the email analysis procedure. All the detected phishing indicators were recorded, and the analysis results were used to classify the email as Phishing, Suspicious, or Safe.

The third step was to classify the email according to the detected phishing indicators and their impact on the business. The classification included the Phishing, Suspicious, and Safe categories. Each category was characterized by the presence of particular phishing indicators and their impact on the business if the email message was opened.

The final step was to prepare the phishing email detection report that included the screenshots of the email messages under analysis, email analysis results, phishing indicators, email classification, business impact, and the recommendations on phishing email prevention. The described methodology is based on ethical principles and allows detecting phishing email attempts by analyzing the email content and following the necessary safety precautions. The analysis of the sample emails demonstrated that the methodology can be used to increase cybersecurity awareness in the case of employees being able to analyze the emails and detect phishing attempts.

Assessment Workflow

The phishing email detection and awareness assessment was performed using a specific procedure to ensure the accuracy of results. Initially, the email samples were selected from public sources as phishing, suspicious, and legitimate emails. All emails were examined in terms of sender, subject, content, links, attachments, and other elements. Different signs of phishing emails were detected and listed. Then, each email was categorized as phishing, suspicious, or safe based on the findings. The potential impact of each email type was defined, and the recommended prevention measures and user awareness considerations were determined. Following that, the assessment results were summarized in a phishing detection and awareness report along with relevant screenshots and other details.



Email Sample Collection

The initial step in the assessment involved gathering a set of public email samples for analysis and phishing awareness purposes. In order to ensure ethical and safe practices, the samples were obtained from reliable GitHub repositories that provide phishing examples for educational, research, and security awareness purposes. In this process, no personal accounts, private information, or organization's emails were accessed or compromised.

Particular attention was paid to emails' subjects and content as well as any attachments and hyperlinks. In total, there were 5 collected email samples. Out of them, 3 samples were related to phishing, one – suspicious, and one – safe. In this assessment, the collected emails were taken as a general reference of email communication to define similarities and variations among phishing emails and those that are not. Additionally, it allowed to demonstrate how the risk level is determined and defined for each particular email messages. In particular, the 3 phishing email samples included one Fake Fax Message, one Fake Voice Message, and one Microsoft's Password Expiration email. In addition, there was one Suspicious Account Activity email and one Safe Account Notification. The phishing-related samples were selected based on the frequency of fake fax and fake voice message types of attacks, as well as password expiration scam, which are common in phishing activities. Moreover, the Suspicious Account Activity email message was included in order to see the difference with Safe Account Notification and determine whether there are all necessary signs for suspicion. Each email message was analyzed in detail in secure environment without clicking on any links or opening attachments.

However, such elements as email address, subject, content, attached files, hyperlinks, branding, and formatting were considered and evaluated. Moreover, each email message's screenshot was taken and included in this document for reference. In this way, the collected set of email messages became the basis for current phishing analysis and awareness assessment. Indeed, the comparison of different types of emails allowed identifying phishing-related similarities and differences between them in order to determine their risk level and provide awareness guidelines.

Email Sample	Category	Purpose
Fake Fax Message	 Phishing	Fake attachment phishing
Fake Voice Message	 Phishing	Malicious voice message attachment
Microsoft Password Expiration	 Phishing	Credential theft attempt
Suspicious Account Activity	 Suspicious	Requires verification before action
Safe Account Notification	 Safe	Legitimate email example

Table 2- Phishing Analysis – 5 email samples categorized as Phishing, Suspicious, and Safe

Phishing Email Analysis

The phishing email analysis was conducted to examine the peculiarities of the presented email samples and determine if they represent phishing attempts or not. The purpose of the analysis was to identify phishing techniques, assess the associated risks, and raise awareness of possible security threats. Since phishing attacks are primarily based on social engineering, the examination of each email was performed to reveal the corresponding manipulation techniques used by attackers. Particular attention was paid to the red flags indicating phishing attempts, as they help recognize deception and protect users from various security threats, such as identity theft, fraud, computer viruses, and unauthorized access to company databases.

Five email samples were analyzed in this report, which were grouped into three categories: Phishing, Suspicious, and Safe. Each category was examined using the same methodology to ensure the reliability of the obtained results. The analysis consisted of multiple steps, including the review of the email header, subject, recipients, body, formatting, hyperlinks, attachments, brand, and language. Particular attention was paid to the presence of phishing red flags, such as sender's email address, generic greetings, alarming language, suspicious hyperlinks, unexpected attachments, requests for sensitive information, grammatical errors, inconsistent brand, and others.

The analysis was conducted in a secure environment to ensure the safety of the researcher. No suspicious links were opened or attachments were opened. The examination of each email was performed using public sources of information and phishing training resources. Multiple screenshots were taken during the analysis to support the findings and ensure the accuracy of the results.

Based on the results of the analysis, the examined email samples were divided into three categories: Phishing, Suspicious, and Safe. The categorization was determined by the presence of phishing red flags and the associated risks identified during the analysis. In addition, the analysis revealed the techniques used by attackers to manipulate the victims and achieve the attack's goals. The findings of the analysis are presented in detail in the following section, which describes each email sample and the corresponding findings. The analysis provides a better understanding of the methods used by attackers and contributes to the increased awareness of email-based security threats.

Email 1 – Fake Fax Message

Email Type: Fake Fax Notification

Risk Level: ● Phishing

Analysis

The first email, which was selected for analysis, is classified as Fake Fax Message type. It can be seen that the email pretends to be a regular fax notification, in order to mislead the addressee and lure him or her into downloading or opening the attachment. This is a classic phishing example, intended to harvest personal data by distributing malware and allowing the hacker to gain unauthorized access to the target's computer or network. Firstly, the given email has a suspicious "from" address; hence, it is better not to open such types of messages. Trustworthy fax services provide the ability to use the company's verified email addresses. Secondly, the subject line seems to be attractive enough, in order to arouse the victim's curiosity and encourage him or her to open the attachment. Moreover, in case of the unexpected arrival of fax notification, one should be careful and not open the dubious file. Cyber criminals use various phishing techniques in order to trick users into opening attachments containing malware or phishing links. It is better to delete such types of emails or reply to them, in order to verify their authenticity. Nevertheless, the above mentioned email has several suspicious signs, such as lack of organization's information, suspicious "from" field, unexpected arrival of fax notification, untrustworthy subject line. In addition, the email lacks general organization information, such as company name, address, phone number, etc. It is a significant red flag, which indicates that the message may contain phishing attempts or malware.

Phishing Email Indicators:

- Suspicious sender's email address
- Unexpected fax notification
- Unexpected attachment
- Lack of organization details
- Social engineering
- Malware distribution methods

Risk Assessment Summary

The email has been classified as phishing due to the presence of multiple phishing indicators. The email address is suspicious and seems to be fake, in order to mislead the victim. Moreover, the "unexpected fax notification" and "unexpected attachment" indicate that the email may contain malware or phishing links.

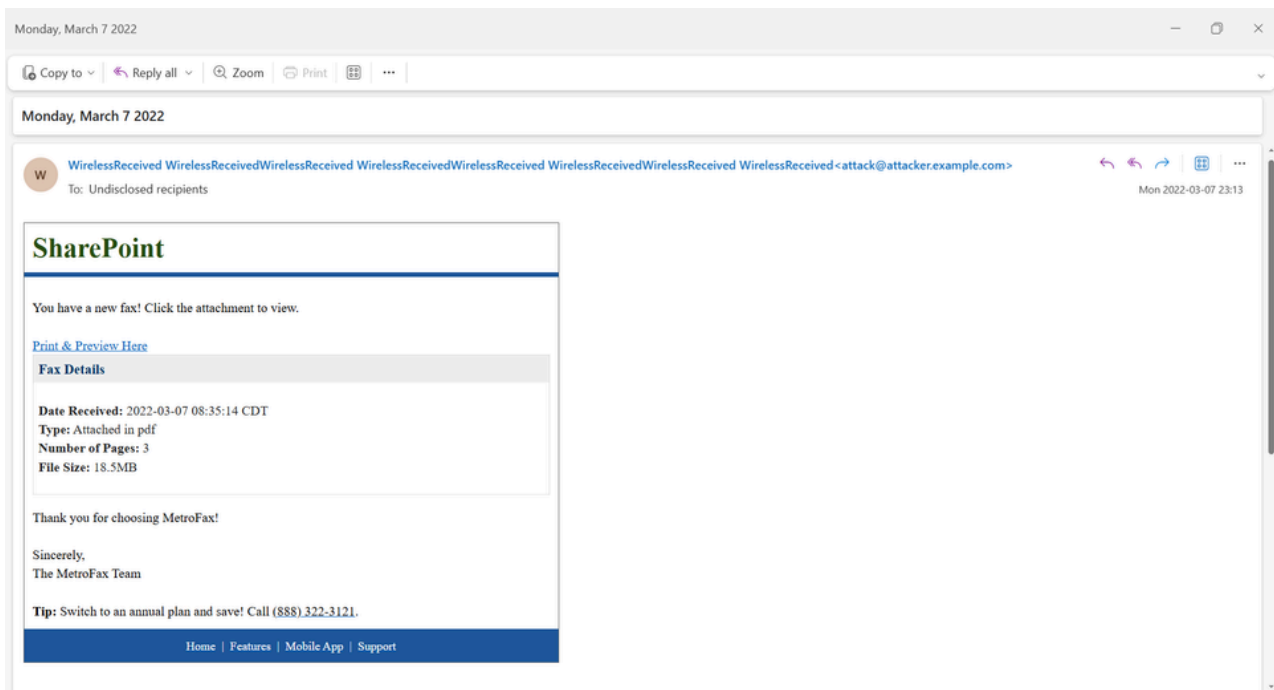


Figure 1- Fake Fax phishing email sample.

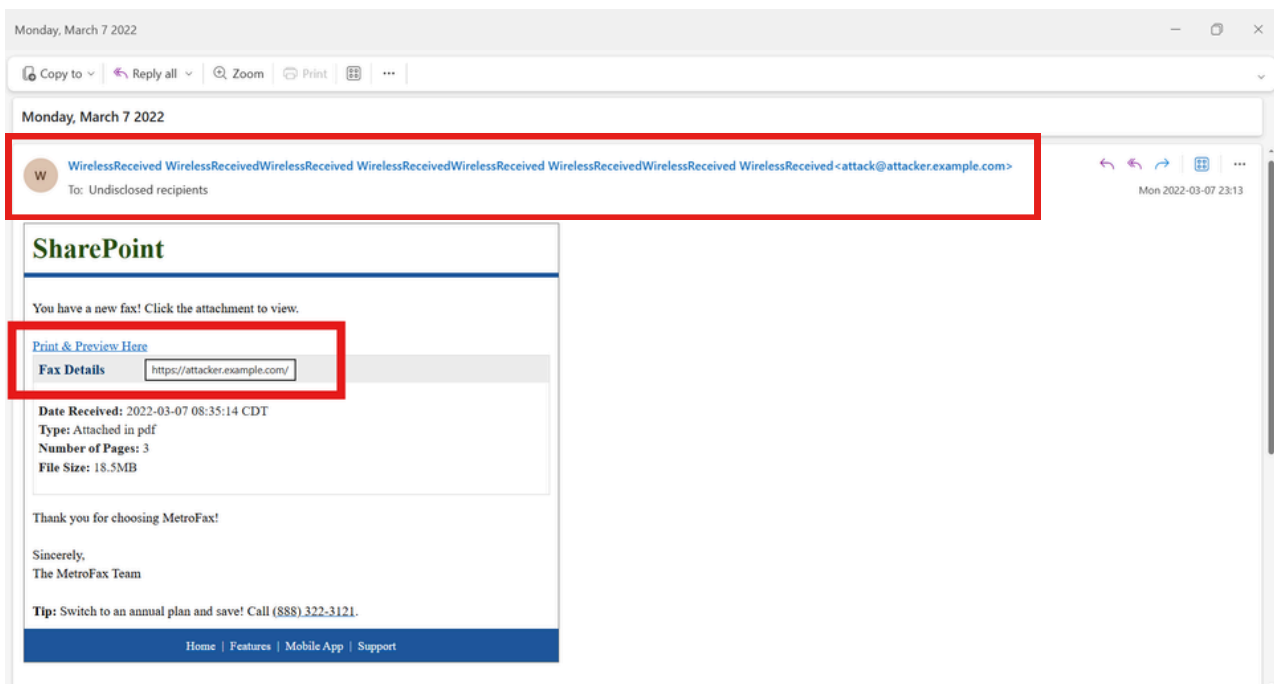


Figure 2- Phishing indicators identified in the Fake Fax email.

Email 2 – Fake Voice Message

Email Type: Voice Message Phishing Email

Risk Classification:  Phishing

Analysis

The second email is created in such a way to remind the user of a new voice message. In addition, the user is asked to download the attachment to access the voicemail. It is a popular phishing method because people tend to ignore the spam and automatically open the file because it was sent as a voicemail notification.

However, during the inspection, multiple suspicious elements were detected that are typically found in phishing messages. Firstly, the email address of the sender is not related to any voicemail service; therefore, it is necessary to delete such letters from the inbox. The fraudsters used the Voicemail service not to notify the user of a new voicemail but to lure the victim into an attachment. The letter was created in such a way that it looked like an official notification from Voicemail. For example, it includes a caller ID, airtime, and the time of the message. However, the email contains psychological tricks that serve as phishing indicators. The main psychological trick is the sense of urgency. The user may feel that a business call is waiting for them, so they have to open the attachment to check what is in the voicemail. Additionally, the sender has used a fake voicemail notification to attract the attention of the victim.

The attachment to the message is the most dangerous part of the email. Such type of phishing is dangerous because voices are not typically sent by email, so voice messages are more likely to be spread through a Voicemail app. It can be concluded that the suspicious file may contain malware that can, for example, encrypt the data on the device or steal the personal information of the victim.

Phishing Indicators

- Suspicious sender's domain
- Fake voicemail notification
- Unexpected attachment
- Sense of urgency
- Psychological tricks
- Malware distribution possibility

Risk Assessment

The given email is classified as phishing due to a number of suspicious elements and the potential danger it can cause. For example, the attachment can provide access to the malware distribution service. As a result, the email can be considered dangerous, and it is advisable not to interact with the suspicious elements of the letter and delete the message from the inbox.

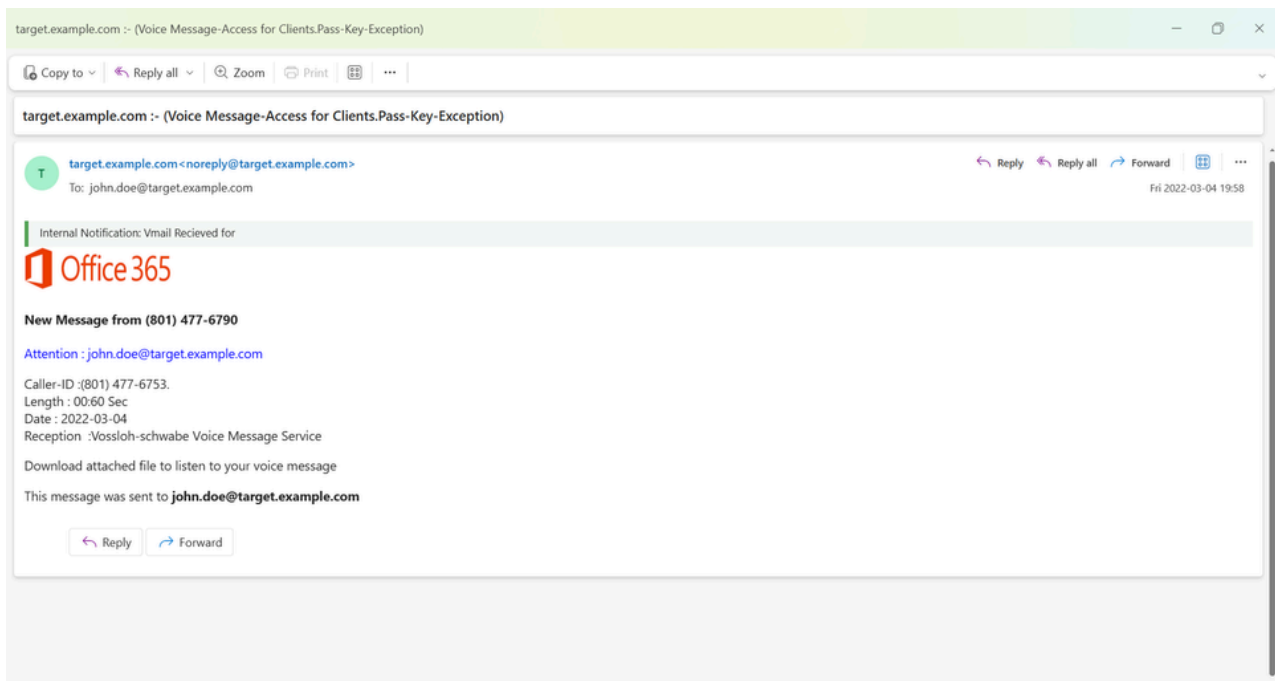


Figure 3- Fake Voice Message phishing email sample.

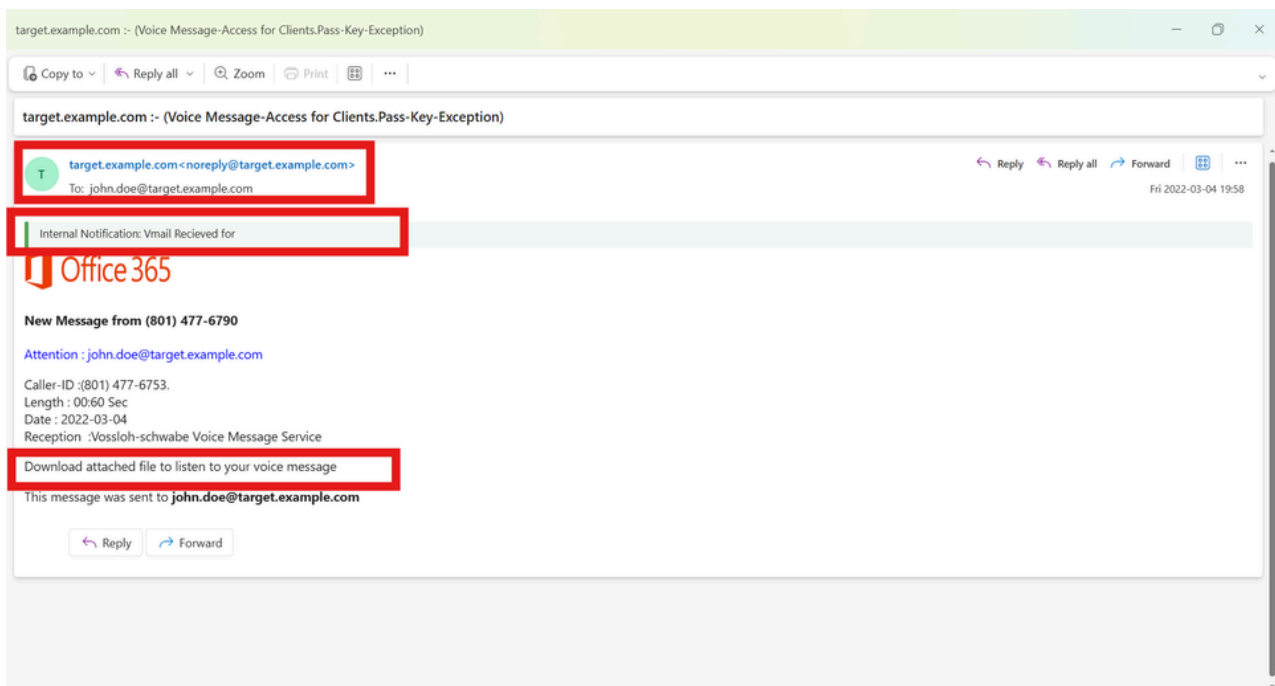


Figure 4- Phishing indicators identified in the Fake Voice Message email.

Email 3 – Microsoft Password Expiration

Email Type: Password Expiration Scam Email

Risk Category:  Phishing

Analysis

The email is an example of phishing scheme, where scammers impersonate Microsoft to mislead the victim into believing that their password has expired. The email asks to keep the same password by clicking the button below. Password phishing scams are the most common type of phishing targeting Microsoft 365, Outlook, and Office accounts.

While the email features the Microsoft logo and brand, there are a number of signs that indicate that this is not an official message. First, the email sender is not genuine since it does not appear from the Microsoft domain. Moreover, the salutation is not personalized but generic instead, which should raise suspicion right away.

The email exploits the panic button as many phishing campaigns do in an attempt to trick the victim into taking immediate action without thinking. The button directs the user to a phishing website that will attempt to steal the user's credentials by luring them into entering their login information on a fake page. In addition, the email's subject line and body suggest that the user's account security is at risk, which can trigger panic.

A legitimate email sent by Microsoft will always come from an official Microsoft domain and provide account management options. It should also be possible to confirm the request by logging into the Microsoft portal and checking the request status directly.

Phishing Indicators

- Microsoft impersonation
- Fake branding
- Suspicious sender
- Password expiration scare tactic
- Credential harvesting
- Urgent language

Risk Assessment

The email is phishing since it is designed to harvest the user's credentials by impersonating Microsoft.

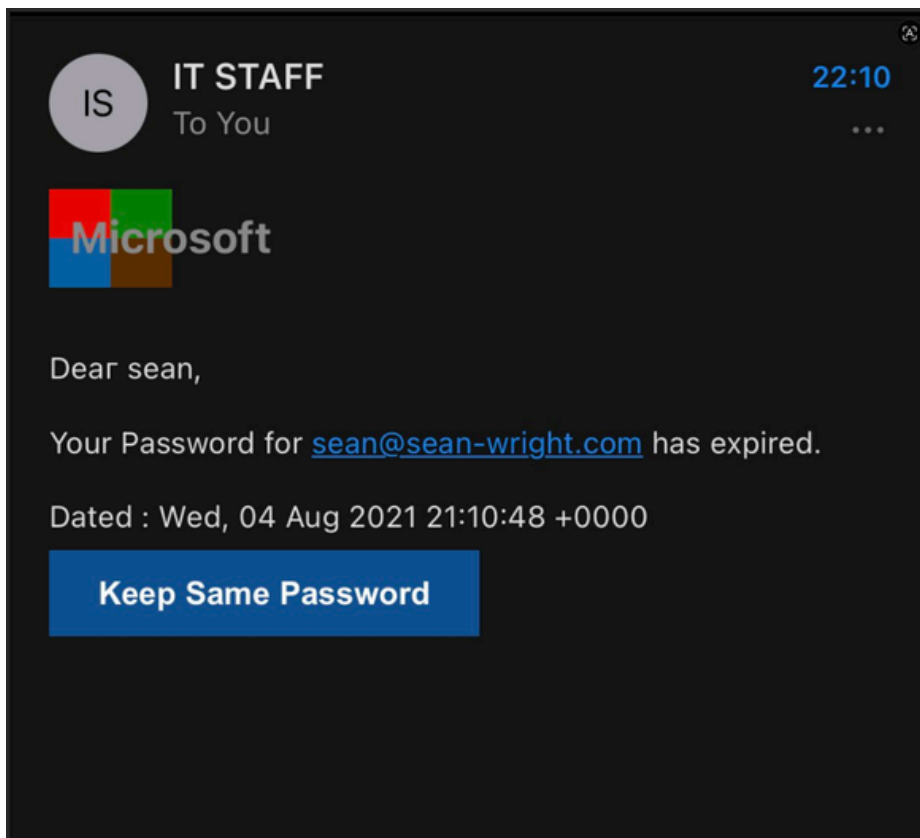


Figure 5- Microsoft Password Expiration phishing email sample.

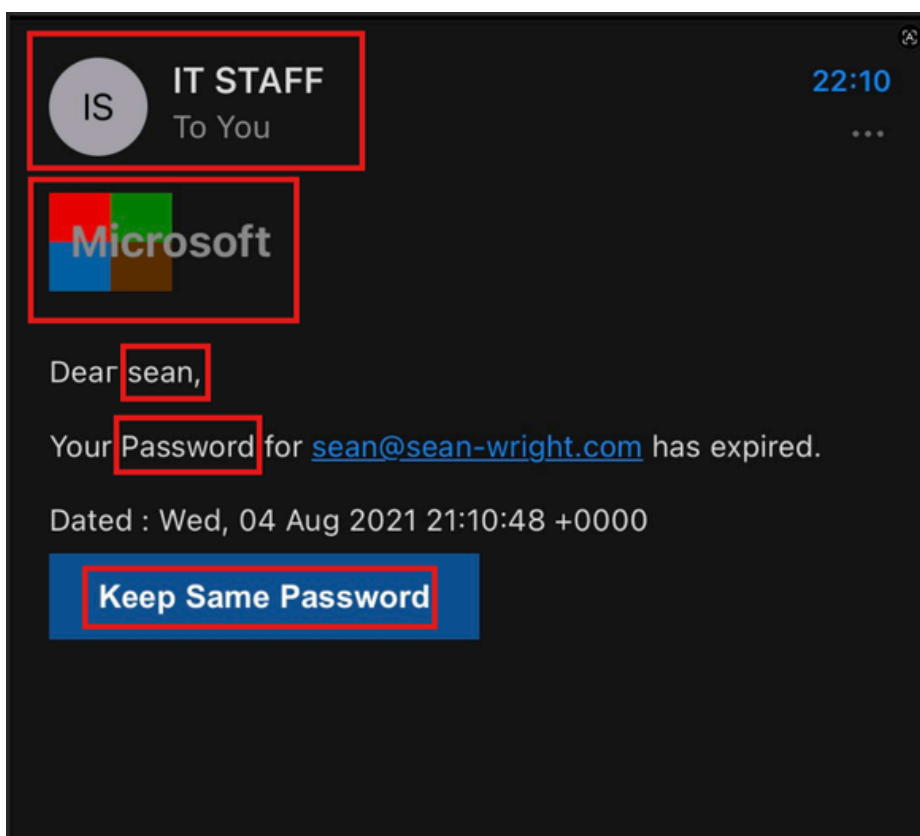


Figure 6- Phishing indicators identified in the Microsoft Password Expiration email.

Email 4 – Suspicious Account Activity

Email Type: Account Activity Notification

Risk Classification:  Suspicious

Analysis

This email warns the recipient about possible account activity and asks to check the account information. Unlike previous phishing emails, this one does not ask to type in the password or money details. The message encourages contacting the organization's IT Help Desk by the provided link to the service desk if something seems suspicious. The language of the email is not threatening and does not force taking any immediate action. There are no words that imply that the recipient's account is at risk or will be closed unless specific requirements are met. In addition, there are no attachments or obvious phishing links. At the same time, the sender's domain is unknown since it cannot be verified as a legitimate address of the company or service. The email is suspicious, and it is better not to take any actions before confirming this message as safe.

For more information about an account activity, it is better to directly contact the company and ask whether this account really requires any authorization or not. In other words, since it is unclear who exactly has sent this message, it is safer to double-check all details before changing anything. Overall, this email cannot be considered as spam or phishing due to some legitimate aspects (mentioning real concerns, providing reliable contact information for verification, professional language). However, there are suspicious elements (unknown address, possible threats to the account) that require individual examination.

Suspicious Indicators

- Unknown sender domain
- Unexpected security notification
- Identity requires verification
- Positive Indicators
- No credential request
- No malicious attachment
- No urgent threats
- Advises contacting official IT support
- Professional language

Risk Assessment

This email is marked as Suspicious because it has a moderate amount of threat indicators, and it should be individually verified before trusting the sender.

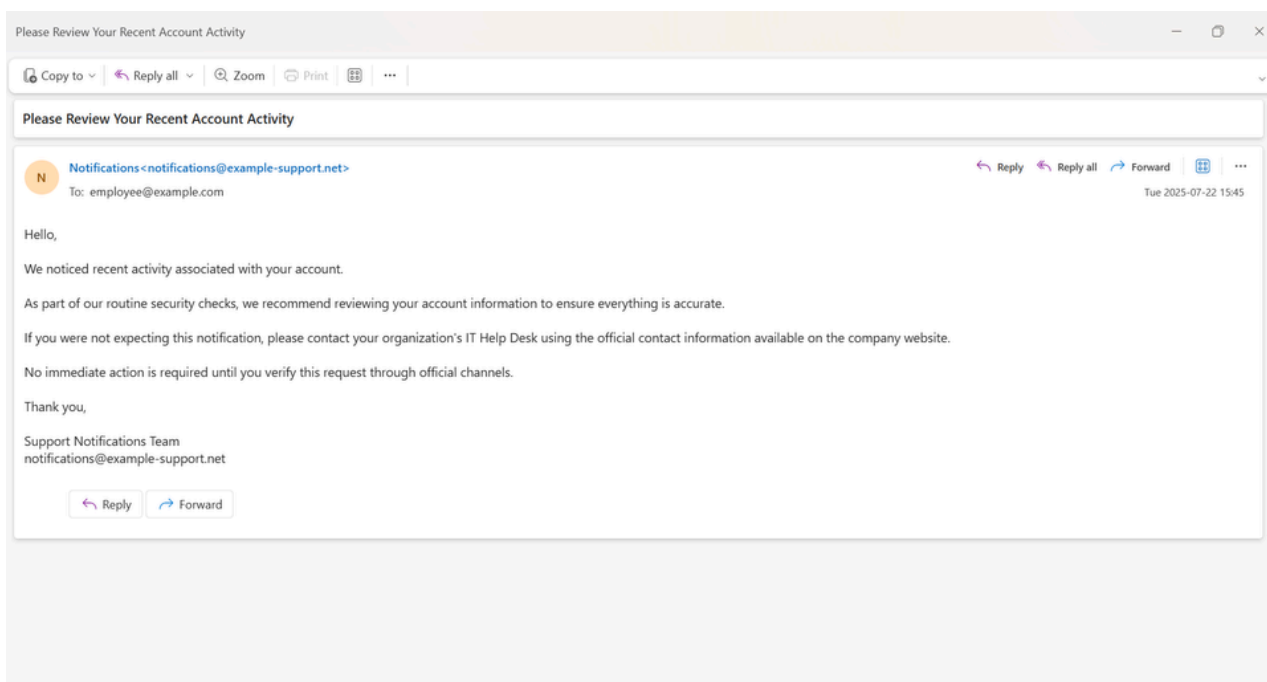


Figure 7- Suspicious Account Activity email sample.

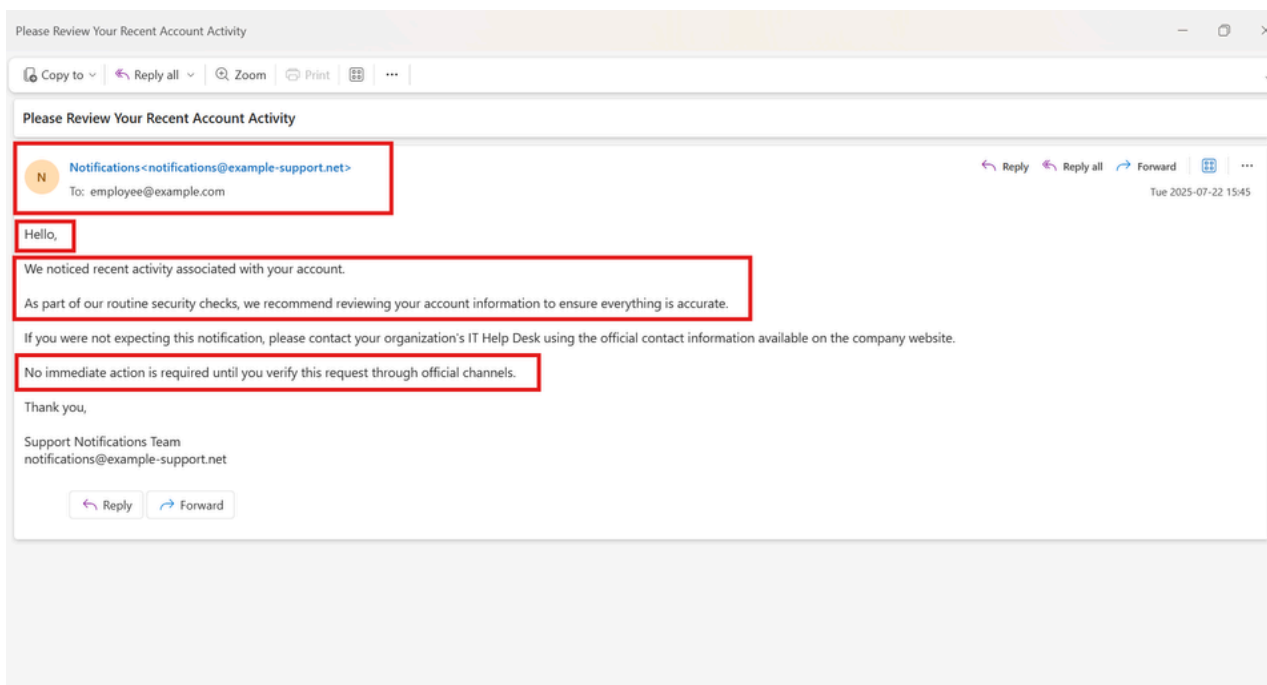


Figure 8- Suspicious indicators identified in the Account Activity email.

Email 5 – Safe Account Notification

Email Type: Legitimate Email

Risk Classification:  Safe

Analysis

The last email under consideration is a legitimate one. The main reason is that this message comes from a reliable source, uses appropriate language, and does not contain any suspicious attachments or hyperlinks. This email was included in the list to provide an example of a legitimate message and highlight the differences between phishing and legitimate emails.

Unlike phishing emails, this message does not attempt to induce panic among users. The email address of the sender is reliable, and the content of the message does not ask to provide any sensitive information. In addition, the email has a legitimate purpose because it informs the recipient about the status of the message. Moreover, no confidential information such as passwords, banking details, or one-time codes is requested. This communication tool also lacks suspicious links and attachments. If the links were legitimate, they would lead to the official website of the company, allowing the recipient to verify the authenticity of the message independently. Furthermore, this email is properly formatted, and no spelling or grammar mistakes could be observed. Considering the information above, it is possible to conclude that this email is legitimate and safe.

Safe Indicators

- Verified sender
- Professional formatting
- Appropriate organization branding
- No request for credentials
- No suspicious links
- No unexpected attachments
- Legitimate communication purpose

Risk Assessment

The email has been assessed as Safe because no phishing indicators have been identified. This message lacks suspicious links, attachments, and language and has a legitimate purpose. Therefore, this email poses a low risk for the user.

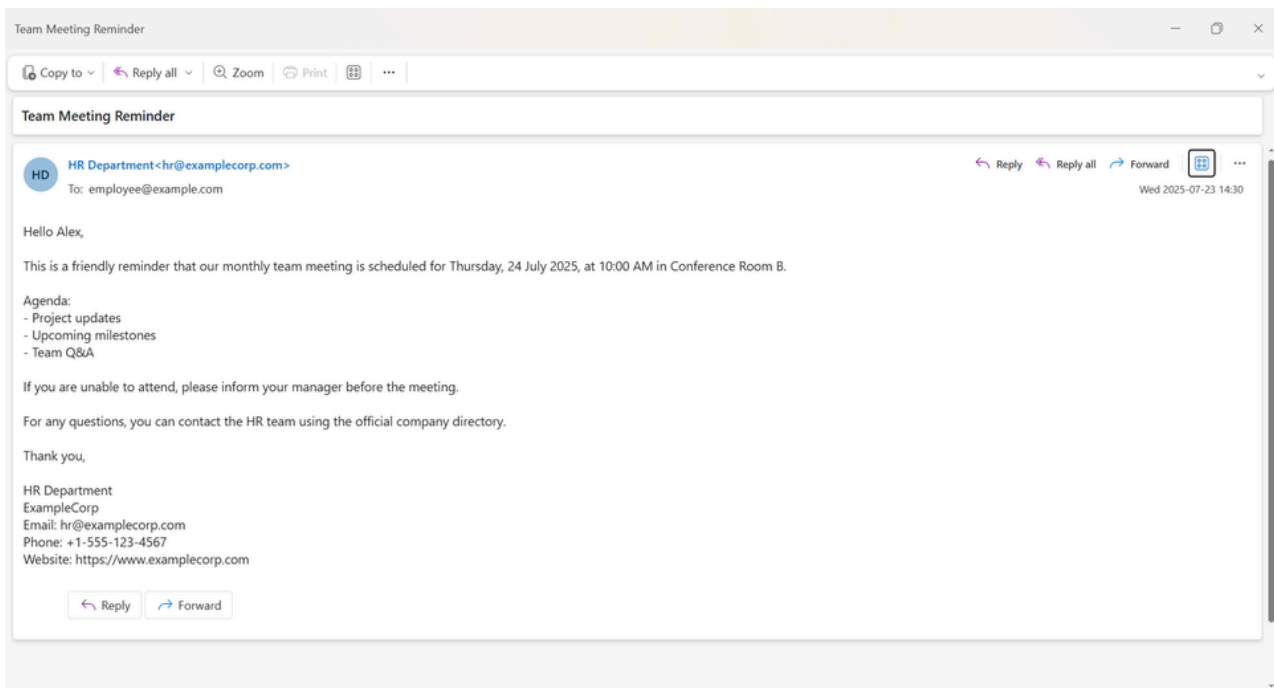


Figure 9- Safe Account Notification email sample.

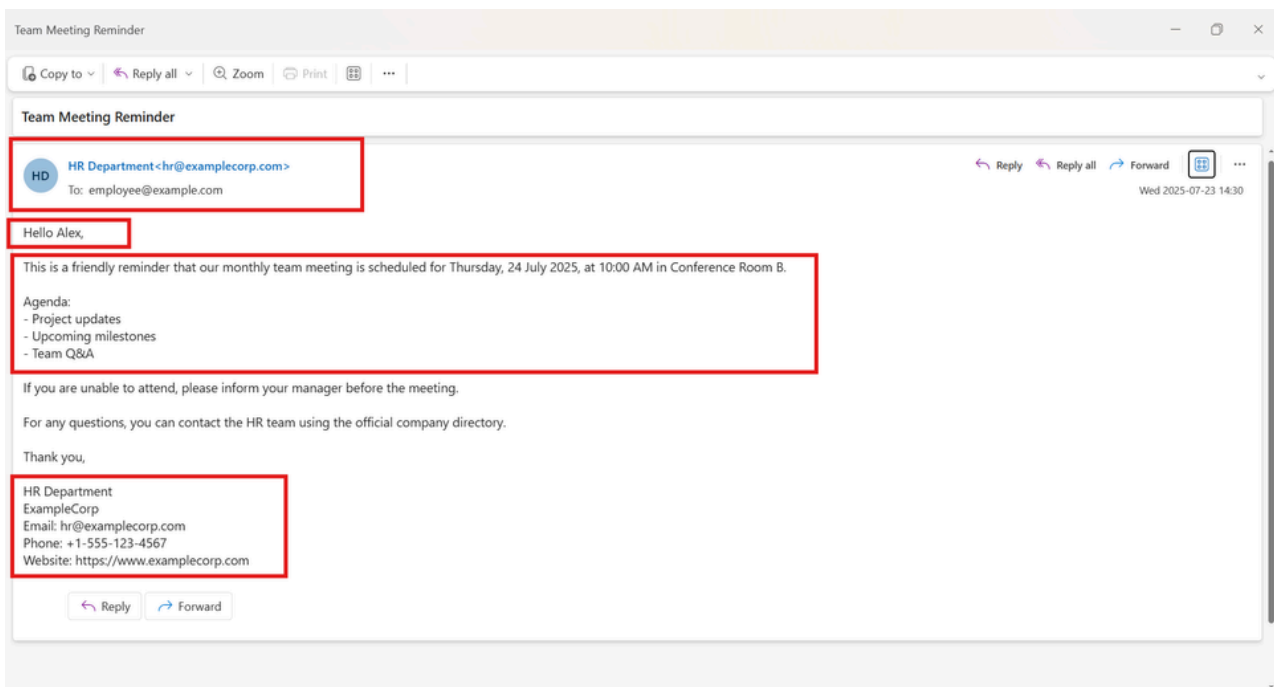


Figure 10- Safe email characteristics identified in the Account Notification email.

Phishing Indicators Identified

Phishing Indicator	Fake Fax	Voice Message	Microsoft Password	Suspicious Email	Safe Email
Suspicious Sender Domain	✓	✓	✓	✓	X
Generic Greeting	✓	✓	✓	X	X
Urgent Language	✓	✓	✓	X	X
Suspicious Link	X	X	✓	X	X
Unexpected Attachment	✓	✓	X	X	X
Requests Credentials	X	X	✓	X	X
Professional Branding	X	X	Partial	Partial	✓
Risk Level	●	●	●	●	●

Table 3- Comparing phishing indicators across email samples

The examination of the provided email samples identified several phishing indicators associated with the attempts of cyber attackers to mislead the recipients. The phishing indicators can be helpful for users to assess suspicious emails and avoid becoming victims of cybercrime.

The first strong phishing indicator detected in multiple email samples was the use of spoofed email addresses from organizations. In particular, several phishing emails were disguised as genuine messages; however, the from addresses were different from the actual organization's domains. One of the phishing indicators also included the use of urgent language or scare tactics, such as informing the recipient that his or her password will expire, the account has been disabled, there are new voicemails, or faxes waiting for review. The use of urgent language and scare tricks compels the recipient to take immediate action without thinking about whether the email request is legitimate or not.

The examination also revealed unexpected attachments and links as a significant phishing indicator. In particular, phishing emails often included attachments disguised as invoices, voice messages, or fax documents that may contain viruses or other types of malicious software. Moreover, many phishing emails contained hyperlinks that directed to websites designed to look like official web portals but are fake and intend to steal user's personal confidential information, including passwords, social security numbers, and other data.

Other phishing indicators included generic greetings in the email body or requests for disclosing personal information, such as usernames and passwords, the presence of grammar or spelling mistakes, and the use of different domains not associated with the purported sender. Notably, legitimate companies use personalized communication with clients, refrain from asking for the disclosure of personal information, and do not contain multiple spelling or grammar mistakes.

Based on the examination, it is important to note that not all suspicious emails are phishing attempts. Some emails addressed as spam or suspicious may involve legitimate activity, such as the given example in the suspicious email category, which did not contain any phishing indicators. At the same time, the examined legitimate email sample demonstrates no signs of phishing attempts, including the presence of a verified sender, clear objective of the message, and its absence of suspicious attachments or hyperlinks.

Overall, the phishing indicators are crucial to ensure cybersecurity and protect users from phishing email scams. It is also vital for users to follow several essential guidelines to minimize the possibility of becoming a victim of cybercrime, such as double-checking the sender before responding or providing any information, avoiding clicking on any links and attachments in case of being suspicious. Furthermore, it is vital to report such cases to the relevant organizations to improve overall email security and address potential weaknesses.

Phishing Indicator	Observed	Impact
Suspicious or spoofed sender domain	✓	Impersonates trusted organizations
Urgent or fear-based language	✓	Pressures users into quick actions
Unexpected attachments	✓	May deliver malware or ransomware
Suspicious hyperlinks	✓	Can redirect users to fake login pages
Generic greetings	✓	Indicates mass phishing campaigns
Request for sensitive information	✓	Attempts to steal credentials or personal data
Inconsistent branding or formatting	✓	Suggests the email is not genuine
Grammar and spelling errors	✓	Common in fraudulent emails
Unverified sender identity	✓	Requires independent verification
Professional branding and verified domain (Safe email)	✓	Indicates a legitimate communication

Table 4- Identifying phishing indicators in suspicious emails

Risk Classification

The process of risk classification is critical to the analysis of phishing emails since it allows the identification of the level of threat posed by each message. In my assessment, I used the indicators of phishing identified in the previous section to classify each email. The classification was based on the information found in the email, including the sender's address, the subject, the body, links, attachments, request for sensitive information, urgency, branding, and overall authenticity of the email.

The classification process was based on the presence of the indicators of phishing identified in the previous step. According to the presence and number of indicators, I divided the emails into three categories: Phishing, Suspicious, and Safe. Each category was defined by the following criteria:

1. Phishing emails contain multiple indicators and pose the most danger to the organization.
2. Suspicious emails have several indicators but lack substantial evidence to be classified as phishing emails.
3. Safe emails contain minor indicators and do not pose any risk.

Phishing emails were given the highest priority since they represent the primary threat to the organization. They usually contain the most significant number of indicators of phishing activity. For instance, they can contain several red flags and attachments that can compromise the computer if opened. On the other hand, I classified suspicious emails as those that have some indicators of phishing but are not substantial enough to take any action immediately. Finally, safe emails had some minor indicators of phishing that lacked sufficient evidence to support them. Based on my analysis, out of the five emails, three fall under the phishing category, one suspicious, and one safe. The three phishing emails contained multiple indicators that showed that they aimed to trick the user into providing sensitive information by using social engineering techniques. The other email that I classified as suspicious had no indicators of phishing except the address of the sender, which was not recognized. The last email was safe because it contained no indicators of phishing and had standard organizational email features.

The risk classification of phishing emails is critical to the overall email security of an organization. Not only does it enhance the security awareness of the employees, but it also helps the organization respond to phishing emails appropriately. In this case, the classification helped me identify phishing emails, including those that were disguised as legitimate emails. This classification process is vital to enhancing email security since it reduces the risk of being tricked into providing sensitive information to cyber attackers. As a result, organizations should conduct routine email risk assessments and provide continuous security awareness training to employees.

Email Sample	Risk Level	Reason for Classification
Fake Fax Message	● Phishing	Suspicious sender, unexpected attachment, fake notification, possible malware delivery.
Fake Voice Message	● Phishing	Fake voicemail notification, unsolicited attachment, social engineering, suspicious sender.
Microsoft Password Expiration	● Phishing	Microsoft impersonation, urgent password reset request, credential harvesting attempt.
Suspicious Account Activity	● Suspicious	Unknown sender domain, no credential request, requires independent verification.
Safe Account Notification	● Safe	Verified sender, professional communication, no malicious links or suspicious requests.

Table 5- Email Analysis Summary – 3 phishing, 1 suspicious, and 1 safe email classified

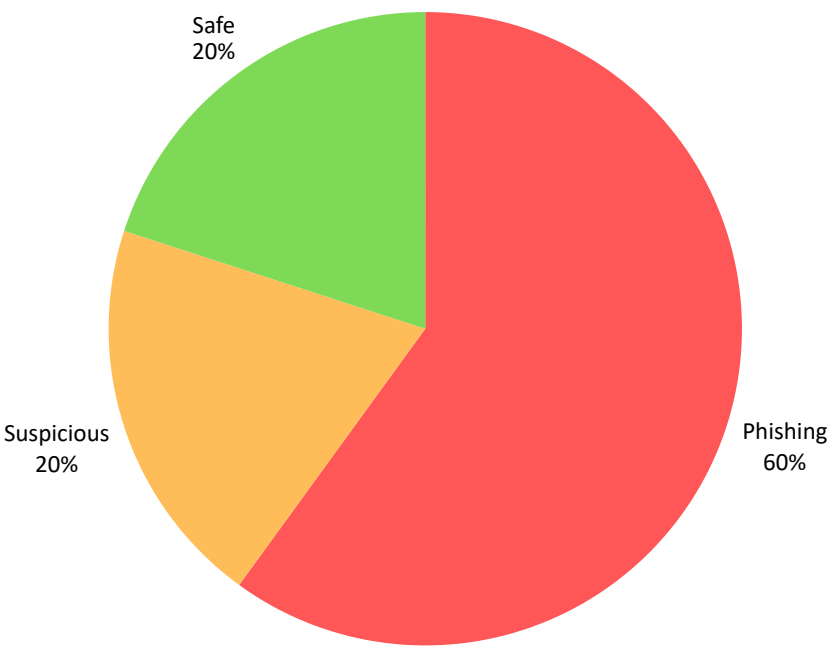


Figure 11- Distribution of analyzed emails by category

Business Impact

Phishing attacks can have serious consequences for both individuals and organizations, and, in this regard, they are one of the most dangerous security threats. Unlike attacks on software, phishing involves human exploitation and manipulation. A single successful phishing attempt can lead to financial loss, disruption of business processes, and reputational damage. Thus, it is important to understand what impact phishing attacks can have on the business in order to build an effective security awareness program.

Firstly, phishing emails can compromise the confidentiality of information and lead to the theft of credentials. In particular, attackers can use phishing emails to steal usernames and passwords and gain unauthorized access to company systems and data. In addition, phishing can lead to the hijacking of business accounts and the theft of customer information.

Secondly, phishing emails can contain malware or ransomware. Attackers can mask malware as an invoice or voice message attachment, the opening of which will allow hackers to gain control of the victim's computer or hold it hostage. As a result of such attacks, businesses may suffer financial losses, experience business disruption, and have to spend time and money to recover data.

Thirdly, phishing can lead to direct financial loss for the company. In particular, phishing can cause fraudulent transfers of funds, hijacking of business communications, and theft of financial information. In addition, the company may have to spend money on legal fines, incident response, and cybersecurity improvements.

Fourthly, phishing can damage the company's reputation and reduce customer trust. In particular, clients and partners are likely to lose confidence in the company's ability to protect their data and provide reliable communication. As a result, the company may suffer reputational damage and loss of potential customers.

The analysis presented in the report demonstrates that phishing emails can have a considerable impact on business. In this regard, it is important to ensure that employees are aware of the dangers of phishing emails. In particular, organizations need to ensure that employees know how to behave in case of phishing emails, report suspicious messages, and take phishing threats seriously. In addition, it is important to educate employees about the basics of cybersecurity and information security to minimize the risks of phishing attacks and protect the organization from threats.

Prevention and Awareness Guidelines

Preventing phishing attacks involves technical and human factors. Since phishing relies heavily on deception and manipulation, it is vital to raise awareness among employees regarding the importance of email security. Users should be encouraged to question the authenticity of every email message and pay close attention to any suspicious activities. The following are email security tips that can help minimize phishing attacks in an organization.

Whenever a user gets an email, it is essential to check the sender's email address. Most spammers use fake addresses that may slightly differ from the original ones. Therefore, users should look out for any irregularities in the email address before replying, responding, or clicking on any attachment or link. If the sender's identity is not recognizable, it is advisable not to take any action but cross-check the email address with the official website.

When interacting with emails, it is crucial to avoid clicking on any suspicious links or downloading unexpected or unwanted attachments, particularly if the email seems undesirable or uninvited. Users should also be wary of any email that uses urgent actions to trick them into disclosing sensitive information. Whenever a suspicious link appears in the email message, it is advisable to avoid clicking by hovering the cursor over the link to view the actual website. Similarly, one should not open the attachments unless their authenticity is undisputed. Most phishing emails trap victims after clicking on the links or opening the attachments. Phishers normally disguise as reliable senders using invoice or voice message files to lure victims into opening the attachments.

Whenever an email request is made to provide vital information, including updating passwords or providing financial credentials, it is essential to confirm the request using an independent source. Users should not reply to the email address provided in the phishing email but use an alternate and reliable contact information to cross-verify the request. It is a good practice to contact the organization's IT department to confirm such emails or report them if no suspicion arises.

The most fundamental step in minimizing the success of phishing attempts is by using MFA. MFA requires users to enter different verification codes to confirm their identity. Even if the spammer managed to acquire the login credentials of an organization, Multi-Factor Authentication (MFA) will prevent unauthorized access to the account.

Conducting regular training sessions on phishing awareness can help users understand the likely phishing attempts, indicators of phishing emails, and how to respond when dealing with emails. In addition, it is essential to develop a responsible disclosure policy that allows users to report suspected phishing attempts by sending them to the relevant authorities, where they can be taken down or investigated. Ideally, users should always assume that emails from unknown senders are mostly spam unless the sender's identity is authenticated.

Besides training, organizations should invest in email security solutions to bolster the email security posture. Advanced spam filters and phishing tools can help filter spam emails before they land in the recipient's inbox. Email security solutions include secure messaging, email encryption, email authentication standards like SPF, DKIM, and DMARC, and antivirus programs. Together with training, technical safeguards like email filtering can help organizations prevent and detect phishing emails, track the compromised users, and collect relevant evidence to prosecute the offenders.



Employee Do's and Don'ts

Employees are the best defense mechanism in an organization against phishing and other email scams. Organizations have implemented technical controls to protect their systems, networks, and data. These include secure email servers, firewalls, anti-virus software, and multifactor authentication, among others. However, criminals are more likely to target the weakest link in the security chain, which is often the human factor. This is because most data breaches and cyber-attack incidents stem from human error. Thus, simply clicking on a spam link or opening an infected attachment can lead to password compromise and illegal access to confidential company information and finances.

The above facts highlight the need to increase employee cybersecurity awareness. Every staff member, including those untrained in IT, should learn how to identify spam emails and know what to do if they receive one. It is always advisable to be extra careful when dealing with emails requesting sensitive information. Being wary of such emails can prevent the organization from experiencing a huge financial and data disaster.

Employees should also know how to report suspicious emails and avoid acting on them. The company's cybersecurity team can act fast to counter such an attack, thus minimizing the damage caused. Reports of spam emails should be regarded as a critical security weakness in the system, requiring immediate intervention. Cybersecurity awareness sessions and phishing simulations are vital in helping employees identify spam emails. Additionally, regular updates on the latest cyber threats and attack methods can help boost an organization's security intelligence.

The Do's and Don'ts of Spam Emails guide workers on the best practices when handling emails, securing company data, and protecting customer information. Everyone in the organization should take cybersecurity awareness seriously and uphold the best practices for the protection of all stakeholders. By observing the Do's and Don'ts of Spam Emails, organizations can create a safe working environment, which ultimately enhances productivity.

✔ DO's	✗ DON'Ts
Verify sender identity	Click suspicious links
Check URLs before clicking	Open unknown attachments
Report suspicious emails	Share passwords or OTPs
Enable Multi-Factor Authentication	Ignore security warnings
Confirm requests through official channels	Trust emails without verification
Keep software updated	Respond to urgent financial requests
Attend security awareness training	Forward suspicious emails
Delete phishing emails after reporting	Disable antivirus or spam filters

Table 6- Phishing Prevention: Do's vs Don'ts

✓ Employee Do's

- Verify the sender's email address before opening an email or responding to any request.
- Inspect links carefully by hovering over them to confirm they lead to legitimate websites.
- Open attachments only if they are expected and received from a trusted source.
- Use Multi-Factor Authentication (MFA) whenever it is available to add an extra layer of account security.
- Report suspicious emails immediately to the IT or Information Security team using the organization's reporting procedure.
- Confirm sensitive requests such as password resets, payment requests, or account updates through official communication channels.
- Keep software, browsers, and antivirus applications updated to protect against known security threats.
- Participate in regular cybersecurity awareness training to stay informed about the latest phishing techniques and best practices.
- Delete phishing emails after reporting them to prevent accidental interaction.
- Remain cautious whenever an email requests confidential information, financial details, or urgent action.

✗ Employee Don'ts

- Do not click on suspicious links or unfamiliar URLs received through email.
- Do not download or open unexpected attachments, especially ZIP, EXE, HTML, or macro-enabled Office documents.
- Do not share passwords, One-Time Passwords (OTPs), PINs, or confidential business information through email.
- Do not trust emails solely because they use company logos or branding; always verify the sender's domain.
- Do not respond to emails requesting immediate financial transactions or account verification without independent confirmation.
- Do not ignore spelling mistakes, grammatical errors, or unusual formatting, as these can be signs of phishing.
- Do not forward suspicious emails to colleagues, as this may spread the threat.
- Do not disable security features such as spam filters, antivirus software, or browser security warnings.
- Do not assume an email is legitimate simply because it appears to come from a familiar organization.
- Do not delay reporting phishing attempts, as early reporting helps protect other employees and the organization.

Conclusion

This Phishing Email Detection & Awareness Assessment was conducted to analyze a number of public email samples, determine common phishing techniques, understand related risks, and raise awareness of the issue. The current assessment was conducted to highlight how cyber-criminals use social engineering to trick their targets into disclosing private information or distributing malware through phishing emails. It was essential to pay attention to the sender's address, email content, attachments, and links, if available, when detecting phishing emails. Each email sample was reviewed individually to examine the detected risks and determine the level of danger using standard assessment tools.

As a result, three email samples were detected as phishing, one suspicious, and one safe. The assessment helped compare the difference between phishing and safe emails, understand why some emails are labeled suspicious, and highlight that not every unsolicited email is dangerous. It was critical to ensure the security of the organization by double-checking every suspicious email and avoiding interactions with phishing emails. The assessment was also conducted to emphasize that even the most advanced and reliable security measures, including email filtering, antivirus, anti-malware, and multi-factor authentication, can be bypassed by social engineering.

Phishing emails remain one of the most dangerous cybersecurity threats that can significantly impact the organization's performance, reputation, and relationships with business partners and customers. Organizations must increase cybersecurity awareness among employees, provide them with essential knowledge about phishing, and ensure they know what to do if a phishing email is detected. The current assessment, along with regular cybersecurity training and up-to-date protective software, can help fulfill the requirements of an organization, remain compliant, and ensure business continuity. The recommendations described in the report, including the best practices for safe email interaction, verification of any suspicious content or links, employee Do's and Don'ts, and key security measures, will benefit all parties involved to prevent phishing attacks, ensure security, and comply with the latest regulations and standards.

In summary, the current phishing email assessment represents an ethical and professional approach to interacting with emails from unknown senders and detecting potential threats to an organization's cybersecurity infrastructure. In this way, it will be possible to remain protected and compliant in the long run.

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